

Machine Learning



Summary



What did we cover in this class?

♦ Supervised Learning

▪ Regression

- Linear Regression
- Polynomial Regression
- Lasso, Ridge, Elastic Net Regularization
- Regression Trees

▪ Classification

- Knn-Classification
- Bayes Classification (Optimal, Naïve)
- Decision Tree Classification (in all its variants)
- SVM
- ANN

▪ Boosting and Extreme Gradient Boosting for Classification and Regression

♦ Unsupervised Learning

▪ Clustering

- K-means
- EM

1. Introduction and kNN
2. Model Evaluation
3. Linear Regression
4. Advanced Regression
5. Optimal, Gaussian, NB Classifier
6. Decision Trees
7. SVM
8. (Artificial Neural Networks)
9. Boosting
10. XGBoost
11. Partitioning Clustering
12. GMM and EM



Summary of Core Concepts

- ♦ Supervised vs. Unsupervised Learning
- ♦ Generalization – Underfitting vs. Overfitting and the Bias-Variance-Tradeoff
- ♦ Regularization – Controlling Overfitting
- ♦ Ensemble Learning
- ♦ Evaluation Metrics and how to compute them
- ♦ Hyperparameters and how to tune them



Butterflies you should be able to recognize



01 Tagpfauenauge



02 Hauchel Bläuling



03 Schachbrett



04 Zitronenfalter



05 Schwalbenschwanz



06 Aurorafalter



07 Kleiner Fuchs



08 Kaisermantel



09 Taubenschwänzchen



10 Trauermantel



11 Admiral



12 C-Falter

**Thank you
for your active
participation**

